

FinTrace: A RAG Assistant with Traceable Evidence for Corporate Financial Reports

Trello: [COSC2669 WIL Project board](#) | **GitHub:** [Private FinTrace repository](#)

Project aim

As an analyst working in financial reporting or investor relations, I want to ask questions about corporate reports and check stated figures against the authoritative source pages so that I can spot inconsistencies quickly and review the supporting evidence before approval.

Scope. FinTrace will answer questions with citations to the report and page used. It will also compare a submitted figure or claim with the relevant disclosure and return one of four outcomes: Supported, Mismatch, Superseded or Unable to verify. The result will show the source and any calculation performed.

Team, roles and contribution

Student ID	Full name	Proposed role and relevant skills	Contribution
S4196173	Ayushi Negi	Technical Project Lead and Integration: architecture, Python, RAG integration, code review and coordination	16.67%
S4188725	Janhavi Maheshwar Ghate	Test Collection and Evaluation: gold questions, retrieval metrics, citation checks and error analysis	16.67%
S4162697	Shriram Varadarajan	Financial Intelligence and Validation: normalisation, calculations, reconciliation, verdict rules and automated tests	16.67%
S4188084	Guruprasad Simimath	RAG and Information Retrieval: chunking, BM25/vector search, embeddings and integration of a local model	16.67%
S4177991	Yash Keswani	Knowledge Base, Research and Data Provenance: report collection, extraction, metadata and version tracking	16.67%
S4196172	Hashini Santhanakrishnan	Prototype, Responsible AI and QA: evidence workflow, usability, human oversight and quality assurance	16.67%
		Total	100%

Progress so far

Team, domain selection and scope. Our group of six is registered as WIL Project 9 and we began by choosing a domain together. We compared four options: consumer health, an RMIT student FAQ, rental repairs and financial reporting. We assessed them against four criteria for a RAG project: access to authoritative public sources, the ability to write questions with known answers, evaluation beyond simple accuracy and clear stakeholder value. We chose financial reporting because it met all four criteria. A stated figure can be checked against its source and the system must also know when to abstain. We then agreed to start with a handful of authoritative corporate reports, evidence linked to page numbers and a limited set of financial metrics.

Roles, infrastructure and coordination. We split the work into six workstreams and matched each one to a member's declared skills. We also drafted and agreed on the aim together. A private GitHub repository and Trello board are in place. The README records our Group ID, members and working aim. We track progress by linking each deliverable to a Trello card and Git commit.

Member contributions in progress. Each member has begun work in their assigned area. Yash is identifying suitable corporate reports and recording their sources and versions. Janhavi is developing the format for test questions and answer keys, including questions that should remain unanswered. Guruprasad is reviewing the reproduced BM25 baseline and planning the dense and hybrid retrieval components. Shriram is defining the supported financial metrics, normalisation rules and verification outcomes. Hashini is planning how evidence will be presented in the prototype and identifying Responsible AI checks. Ayushi completed the preliminary Walert reproduction described below and is coordinating the system design and integration.

Walert reproduction and preliminary results

Reproduction. We inspected the official Walert project and rebuilt a BM25 retrieval baseline from its 120 FAQ passages. We evaluated all 106 released questions and saved reproducible rankings, metrics and input hashes. To compare retrieval strategies at no cost, we recomputed metrics from Walert's supplied BM25, dense and intent run files rather than running the Falcon-7B and Pyserini pipeline again. The results are below.

Retrieval effectiveness (96 answerable questions)

Retrieval run	MRR	nDCG@5	Hit@5
Rebuilt BM25 (our implementation)	0.641	0.419	80.21%
Walert BM25 (supplied run)	0.603	0.455	77.08%
Walert dense (supplied run)	0.685	0.538	71.88%
Walert intent (supplied run, 72/96 returned)	0.573	0.196	57.29%

Safe refusal on out of knowledge base questions (10 questions)

Generative system (Walert supplied)	Unanswered rate
Intent system	80% (8/10)
BM25 + Falcon (context k = 1 to 5)	10-20% (1-2/10)
Dense + Falcon (context k = 1 to 5)	0-10% (0-1/10)

Interpretation: Our rebuilt BM25 closely tracks Walert's supplied BM25 run. The dense run has the strongest graded ranking (nDCG@5 0.538) but the lowest Hit@5, so no single measure is sufficient. This motivates hybrid retrieval and an evaluation using several metrics. Retrieval was also weaker on inferred questions than on known questions. On questions outside the knowledge base, generators using Falcon refused only 0 to 20% of the time. This shows that the model may attempt unsupported answers. FinTrace will therefore evaluate evidence retrieval and safe refusal separately.

Plan for the next three weeks

Week	Work package and owners	Verifiable definition of done
1	Yash: bounded corpus and provenance manifest. Janhavi: questions and answer keys. Guruprasad: stable BM25 baseline.	Fixed set of official reports; chunks linked to page numbers; answerable and unanswerable test questions; saved baseline metrics.
2	Guruprasad: dense and hybrid retrieval with a local Qwen model. Shriram: financial normalisation and calculation engine. Ayushi: interfaces and integration.	BM25, dense and hybrid runs on one test set; unit, period and currency rules tested; answers linked to evidence and generated locally.
3	Hashini: prototype for showing evidence and Responsible AI checks. Janhavi: final evaluation. Ayushi: full system integration and rubric review.	Working Ask and Verify demo; metric comparison, refusal and citation checks; error analysis; reviewed documentation and demo workflow.

Key risks and controls

Risk	Control
PDF tables or units are extracted incorrectly	Limit supported metrics, retain page images and manually validate values.
Unsupported but confident answers	Require retrieved evidence and return Unable to verify when the threshold or source conditions are not met.
Local model latency	Start with a quantised Qwen3 4B model, cap retrieved context and benchmark response time.
Scope or integration drift	Use a small fixed document collection and move additional companies, metrics and features to future work.

Evidence: Walert reproduction results are stored in walert_reproduction/RESULTS.md.

Source: S. Pathiyan Cherumanal et al., Walert, CHIIR 2024, DOI 10.1145/3627508.3638309.

Group Work: Milestone 1: FinTrace

Group ID: WIL Project 9

Contribution Sheet

For each of member in your team, please write your name, student number, contribution percentage to the assignment and your signature. Marks are awarded to the individual team members, according to the contributions made towards the final work. Please submit what percentage each member made to this assignment and submit this sheet in your submission.

The contributions of your group should add up to 100%.

Name	Student ID	Contribution Percentage	Specific Work on Assignment	Signature
Ayushi Negi	S4196173	16.67%	Project setup, Walert BM25 reproduction and report integration	Ayushi Negi
Janhavi Maheshwar Ghate	S4188725	16.67%	Test collection and evaluation planning	Janhavi Ghate
Guruprasad Simimath	S4188084	16.67%	RAG and retrieval development planning	Guruprasad Simimath
Shriram Varadarajan	S4162697	16.67%	Financial validation and testing planning	Shriram Varadarajan
Yash Keswani	S4177991	16.67%	Research on data sources and provenance planning	Yash Keswani
Hashini Santhanakrishnan	S4196172	16.67%	Prototype workflow, Responsible AI and QA planning	Hashini Santhanakrishnan

Note: Mark calculation according to contributions.

Assume the mark for the presentation is **m**,

If the contribution percentages are even (assuming there are **n** members, $1/n$ contribution for each member is considered as even), then each member receives **m** mark;

Otherwise, if the contribution is not even:

Students with the **highest** contribution (e.g., **X%**) receives **m** marks;

For other students, e.g., let another Student who has contribution **Y%** ($Y < X$), his achieved mark will be calculated as $(Y/X) * m$